

The Geometry of Constraints

Adversarial Representations, Fibrational Semantics,
and the Dependency Architecture of Formal Systems

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Abstract

This essay traces a structural continuity running from the empirical phenomenon of adversarial vulnerability in machine learning through the differential geometry of neural representation spaces, and ultimately to the categorical foundations of dependent type theory. Beginning with the observation that adversarial perturbations are not isolated anomalies but geometric features of learned representation manifolds, the essay develops a unified conceptual framework in which adversarial subspaces, Grothendieck fibrations, comprehension categories, and the λ -cube all emerge as manifestations of the same underlying architecture: a geometry of constraints governing how structures vary across contexts and interact across levels of abstraction. The Cox–Bunzel representational similarity framework and Centered Kernel Alignment are interpreted geometrically as instruments for probing the foliation structure of high-dimensional semantic manifolds. The λ -cube is reinterpreted not as a syntactic taxonomy but as the minimal coordinate system for a stratified dependency lattice, whose axes correspond to adjunctions between levels of a formal hierarchy. Comprehension categories are shown to generate the cube’s structure automatically from the minimal data of a fibration equipped with context extension. Transformer attention mechanisms are analysed as dynamic metric tensors inducing context-dependent curvature on the representation manifold. Representation learning is interpreted as a renormalization flow whose fixed points are the semantic foliations whose transverse directions constitute the adversarial attack surface. A Geometric Summary Theorem unifies all of these observations into a single formal statement. The essay concludes that type theory, categorical semantics, information geometry, and adversarial machine learning are all partial descriptions of a single geometry: the geometry of constraint fields defined over spaces of contexts.

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1 Robustness, Regulation, and the Limits of Verification

Modern neural networks achieve predictive performance that would have appeared extraordinary a generation ago, yet they remain fragile in a manner that has no analogue in classical engineering systems. Minor manipulations of input data—manipulations that appear insignificant or even imperceptible to human observers—can produce large, discontinuous changes in model behavior. These adversarial perturbations are not isolated pathological cases. They occupy large, geometrically structured regions of the input space, and their existence is not contingent on implementation details of any particular architecture. They are, as the subsequent analysis will argue, necessary consequences of the geometry of learned representations in high-dimensional spaces.

This fragility has become a matter not merely of theoretical interest but of regulatory consequence. As artificial intelligence systems become embedded in critical infrastructure, financial decision-making, healthcare, and public governance, their resistance to adversarial manipulation becomes a precondition of responsible deployment. The European Union’s AI Act requires organizations to demonstrate robustness and security as core properties of high-risk systems. Similar frameworks are emerging across multiple jurisdictions. Yet these regulatory demands run immediately into a mathematical obstacle: the adversarial attack surface of a deep neural network is combinatorially enormous, and complete verification of robustness with respect to that surface is computationally infeasible. Even verifying robustness within a small perturbation radius around a single input is NP-hard in the general case.

The central tension of adversarial machine learning research is therefore not primarily a tension between safe and unsafe systems but a tension between the theoretical completeness required by rigorous guarantees and the statistical coverage achievable in practice. Organizations cannot enumerate all adversarial inputs. They can only probe the adversarial landscape using surrogate methods and hope that their probes provide representative coverage. Understanding what “representative” means in this context requires understanding the geometry of representation spaces, which in turn opens onto mathematical structures far deeper than the applied setting might initially suggest.

This essay traces that depth. It begins with the applied problem of adversarial transferability and the representational similarity metrics developed to address it, then passes through the differential geometry of transformer attention mechanisms and high-dimensional embedding spaces, and ultimately arrives at

Grothendieck fibrations, comprehension categories, and the λ -cube. The claim developed throughout is that these apparently disparate topics are unified by a single structural insight: complex systems, from neural networks to logical languages, can be understood as layered spaces of states, constraints, and meta-constraints, and the behavior of adversarial perturbations, program evaluation, and logical proof are all instances of motion through such layered spaces.

2 Geometric Thesis and Essay Roadmap

The argument of this essay turns on the recognition that three apparently independent technical domains—adversarial machine learning, dependent type theory, and categorical semantics—share a common geometric skeleton. Stating that identification explicitly at the outset will allow the subsequent sections to be read as coordinate charts on a single underlying structure rather than as a sequence of loosely related technical expositions.

The three structural parallels are as follows.

The first concerns the degeneracy structure of neural embedding manifolds. A trained network defines a representation map $h : X \rightarrow \mathbb{R}^d$ whose Jacobian field determines two complementary distributions on the input tangent bundle: the tangent directions along which representation varies negligibly (the leaves of the semantic equivalence foliation) and the transverse directions along which small displacements produce large output changes (the adversarial subspace). Adversarial vulnerability is the existence of an abundant, wide adversarial cone in the transverse distribution; robustness corresponds to minimizing its width or geodesic accessibility from typical data.

The second concerns the fibrational structure of dependent type theory. A dependent type judgment $\Gamma \vdash A : *$ describes an object in the fiber of a Grothendieck fibration $p : \mathcal{E} \rightarrow \mathcal{B}$ over the context Γ . A term inhabiting a dependent type is a section of the resulting bundle. Substitution is reindexing along base morphisms. Dependent products are right adjoints to reindexing; dependent sums are left adjoints. Computation is flow along constraint manifolds; the subject-reduction theorem is the condition that this flow respects the fibration structure.

The third concerns the categorical structure of comprehension categories and the λ -cube. A comprehension category formalizes the minimal data required for a language to describe objects together with predicates on those objects. From this minimal data the three axes of the λ -cube emerge automatically as the three ways in which the comprehension fibration’s levels may depend on one another. The cube is therefore not a syntactic classification but the dependency skeleton of any

self-describing formal system.

The central thesis of the essay can now be stated precisely.

The degeneracy geometry of neural representation manifolds, the fibration semantics of dependent type theory, and the categorical structure of comprehension categories are all manifestations of the same underlying phenomenon: constraint fields defined over spaces of contexts, whose sections describe coherent global objects, whose transverse directions describe instability, and whose leaf structure encodes the equivalences that the system has learned or been designed to respect.

The essay proceeds as follows. Sections 3–7 develop the geometry of adversarial machine learning, from transferability and CKA through representation manifolds, Jacobian metric structure, and singular foliations. Section 6 analyses transformer attention as a dynamic metric tensor. Section 8 interprets representation maps as fiber bundle sections and then as sections of Grothendieck fibrations. Sections 9–12 develop the type-theoretic material: the λ -cube as a dependency coordinate system, comprehension categories, Grothendieck fibrations, and the adjoint lattice underlying the cube. Sections 13–14 connect these structures to information geometry and homotopy type theory. Section 15 interprets representation learning as a renormalization flow. Section 16 derives adversarial attacks as geodesic escape from foliation leaves. Section 17 states the Geometric Summary Theorem. Section 19 is the conclusion. The appendix provides the explicit categorical construction.

3 Adversarial Transferability and Representational Similarity

The phenomenon of adversarial transferability was among the first indications that adversarial vulnerability is a structural property of learned representations rather than an idiosyncratic flaw of individual architectures. Early work by Goodfellow, Shlens, and Szegedy established that neural networks exhibit large connected regions of input space in which small perturbations produce dramatic changes in output classification. These adversarial examples arise because the learned decision boundary of a network is highly nonlinear yet locally almost linear in high-dimensional regions. A perturbation aligned with the gradient of the loss function can push an input across the boundary with minimal displacement in input space. The fast gradient sign method formalizes this intuition: given a model M with

loss L_M evaluated at input x , the perturbation

$$\delta_M = \epsilon \operatorname{sign}(\nabla_x L_M(x))$$

produces an adversarial example $x + \delta_M$ that is geometrically close to x yet behaviorally distant.

The transferability observation sharpened this understanding considerably. Adversarial examples crafted on one model frequently deceive other models trained on the same task, even when the architectures differ substantially. This cross-model transfer implies that adversarial directions are not idiosyncratic to individual networks but are aligned with structural features of the shared task geometry. An attack transfers from surrogate S to target T when the gradients are sufficiently aligned:

$$\langle \nabla_x L_S(x), \nabla_x L_T(x) \rangle > 0.$$

The question of when this alignment holds, and how to measure it without direct access to the target model, is the practical problem that representational similarity metrics are designed to address.

Centered Kernel Alignment (CKA) offers one principled answer. Given two networks f and g that map a common set of inputs to activation matrices X and Y , CKA measures the similarity between the pairwise relationship structures induced by these activations. Formally, with $K = XX^\top$ and $L = YY^\top$,

$$\text{CKA}(K, L) = \frac{\|L^\top K\|_F^2}{\|K^\top K\|_F \|L^\top L\|_F},$$

where centering is applied to remove mean effects. CKA produces a value in $[0, 1]$ that is invariant to orthogonal transformations and isotropic scaling. High CKA similarity indicates that two networks organize data into similar geometric patterns within their representation spaces. Because adversarial vulnerabilities are closely tied to the geometry of learned representations, models with similar CKA scores are more likely to share adversarial directions.

The Cox–Bunzel framework operationalizes this observation into a testing protocol for organizations required to demonstrate adversarial robustness. The protocol partitions surrogate models into two groups based on their CKA similarity to the target model. The high-similarity group, characterized by CKA scores above approximately 0.55 for convolutional architectures, provides proxy models that are geometrically close to the target; attacks generated on these surrogates are likely to transfer. The low-similarity group, with CKA scores below approximately 0.35, ensures that the testing protocol also explores adversarial regions orthogonal to

those shared with the high-similarity cluster. By combining attacks from both groups, the protocol approximates coverage of the adversarial landscape without attempting an impossible exhaustive search. The Diagonal Box Similarity metric supplements this global picture with layer-by-layer granularity, allowing comparison of local representation structure at different network depths.

This framework converts the theoretical impossibility of complete robustness verification into a tractable statistical testing problem. But to understand why this conversion works—why representational similarity should predict attack transferability—requires understanding the geometry of the representation spaces themselves.

4 The Geometry of Representation Spaces

Neural networks of sufficient capacity learn to map inputs into high-dimensional embedding spaces in which semantic relationships become approximately linear. This approximate linearity is not incidental. It follows from the interaction between the loss landscape, the implicit biases of gradient descent, and the functional form of the network.

The resulting representation spaces are not smooth Riemannian manifolds in any classical sense. A neural network computes a piecewise smooth map from input space to representation space, assembled from compositions of affine maps and nonlinearities. The geometry of this map determines the network’s sensitivity to perturbations. At a point x in input space, the Jacobian

$$J_M(x) = \frac{\partial h_M(x)}{\partial x}$$

describes how small displacements in input space propagate into representation space. When $J_M(x)$ has large singular values along certain directions, small movements in those directions produce large changes in the representation, and consequently in the model’s output. These high-gain directions constitute the adversarial subspace local to x .

The softmax aggregation inherent in transformer attention also produces a distinctive geometric phenomenon: many distinct inputs can collapse onto nearly identical internal representations. The representation space therefore contains broad regions of approximate equivalence—submanifolds along which the network’s internal state remains nearly constant despite large variation in the input.

These regions of approximate equivalence form structures resembling degenerate foliations. A foliation of a manifold is a decomposition into layered submanifolds of lower dimension, called leaves, that fit together without crossing. In

a standard foliation the leaves are smooth and uniform; in the degenerate foliations that appear in neural representation spaces, the dimensionality and smoothness of the leaves vary across the space. Along directions tangent to the leaves, model behavior remains approximately invariant. Along directions transverse to the leaves—directions crossing from one leaf to another—small displacements produce large changes in model output. The adversarial perturbation problem is precisely the problem of finding these transverse directions efficiently.

5 The Metric Structure of Representation Maps

The foliation interpretation of adversarial subspaces can be made precise by examining the differential structure induced by the representation map.

Let $h : X \rightarrow \mathbb{R}^d$ be the representation map of a trained model, where X denotes the input space and $\mathbb{R}^d$ the embedding space. The Jacobian

$$J_h(x) = \frac{\partial h(x)}{\partial x}$$

defines a linear map between tangent spaces

$$J_h(x) : T_x X \longrightarrow T_{h(x)} \mathbb{R}^d.$$

This map induces a pullback metric on the input manifold. If $\langle \cdot, \cdot \rangle$ denotes the standard inner product in $\mathbb{R}^d$, the induced metric on $T_x X$ is

$$g_x(u, v) = \langle J_h(x) u, J_h(x) v \rangle.$$

The geometry relevant for adversarial perturbations is therefore not the Euclidean geometry of input space but the Riemannian geometry defined by this pullback metric. Directions in which the Jacobian has small singular values correspond to directions along which the representation changes minimally; these form approximate tangent spaces to the semantic equivalence leaves. Conversely, directions associated with large singular values correspond to directions in which representation varies rapidly—precisely the directions in which small perturbations produce large changes in network output.

The adversarial structure of the model therefore emerges directly from the spectral geometry of the Jacobian field over the input manifold. Training implicitly factors the representation space into equivalence classes of inputs sharing semantic invariants. Two inputs x and x' lie on the same leaf whenever their downstream

classifications agree: $f(h(x)) = f(h(x'))$. The tangent space to the leaf at x is

$$T_x \mathcal{F} = \ker D(f \circ h)_x,$$

and it is precisely the directions discarded by the Jacobian's low singular values. The adversarial subspace is the orthogonal complement of this kernel under g_x .

6 Transformer Attention as a Dynamic Metric Tensor

Transformer architectures introduce an additional layer of geometric complexity through the attention mechanism. Unlike feedforward networks, where the pull-back metric is fixed by the model's parameters, transformer attention induces a context-dependent metric that varies dynamically with each input.

For a sequence of token embeddings $H = (h_1, h_2, \dots, h_n)$, the attention mechanism computes query, key, and value projections

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V,$$

and produces the transformed representation

$$\text{Attn}(H) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

The attention matrix

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)$$

defines a similarity kernel over the token representations. Each row of A determines how strongly one token representation is influenced by all others in the sequence. The inner product $\langle q_i, k_j \rangle$ measures the interaction strength between tokens i and j , so attention defines a bilinear form over the embedding manifold that varies with the input context. In differential-geometric terms, this is a context-dependent metric tensor

$$g_x : T_x M \times T_x M \longrightarrow \mathbb{R}$$

over the representation manifold M , whose components shift with every new sequence.

This dynamic metric structure explains several empirical properties of transformer models. Tokens whose query-key similarity is high become strongly coupled

under the induced metric, effectively collapsing nearby directions of representation space and producing the semantic equivalence leaves described in the previous section. Small perturbations of the input can alter the query-key similarity matrix, which in turn perturbs the metric tensor itself, amplifying the effect of the perturbation in a way that is absent from purely feedforward architectures. Adversarial attacks on transformers therefore do not merely exploit sensitive directions in a fixed geometry: they exploit the instability of the metric, destabilizing the geometry and thereby widening the adversarial cone in a self-reinforcing manner.

Each input sequence produces a slightly different metric structure. The representation manifold of a transformer is therefore better understood as a bundle of metrics over input space than as a single Riemannian manifold. The fibers of this bundle are the possible instantiated metrics, and the dynamics of attention determine how the section of this bundle—the metric associated to a given input—varies as the input changes. This structure is precisely that of a fiber bundle, and the neural network’s behavior under the attention mechanism is a section of that bundle, exactly as formalized in the categorical treatment developed in later sections.

7 Adversarial Subspaces and Singular Foliations

The foliation picture allows a precise geometric account of why adversarial examples exist and why they are so abundant. Training progressively separates the singular values of the Jacobian into two regimes. Let $\sigma_1(x) \geq \sigma_2(x) \geq \dots$ denote the singular values of $J_h(x)$. After training, one observes empirically that

$$\sigma_1(x), \dots, \sigma_k(x) \gg \sigma_{k+1}(x), \dots$$

The large singular values correspond to directions that remain relevant to the task; the small ones correspond to directions whose variation the network has learned to suppress. This spectral separation produces the foliation. However, the foliation that arises is not smooth in the classical sense.

Definition 7.1 (Singular Foliation). A *singular foliation* on a manifold M is a partition of M into immersed submanifolds—called leaves—whose dimensions may vary across the space.

Neural representation spaces exhibit exactly this structure. Regions of high semantic compression correspond to low-dimensional leaves, where the network has learned to ignore many directions of variation. Regions of semantic ambiguity correspond to higher-dimensional leaves, where multiple directions of variation

influence the representation. Adversarial perturbations arise with particular potency near the boundaries between these regions: at such boundaries the foliation becomes singular, with the leaf dimension changing discontinuously.

The adversarial subspace at x is the transverse bundle $T_x X / T_x \mathcal{F}$. When the foliation is singular, this quotient space changes dimension across the manifold, creating regions where the transverse directions become unusually large. These regions correspond precisely to the unstable decision boundaries observed empirically.

Definition 7.2 (Adversarial Subspace). The *adversarial subspace* at x is the orthogonal complement

$$A_x = (T_x \mathcal{F})^\perp$$

with respect to the pullback metric g_x .

Theorem 7.3 (Adversarial Directions as Transversals). *Let $h : X \rightarrow \mathbb{R}^d$ be a differentiable representation map and let $f \circ h$ be the associated classifier. Adversarial perturbations correspond to vectors in A_x . If $v \in A_x$, then the first-order change in the classifier loss is maximized along v under the metric induced by $J_h(x)$.*

Proof. Let $L(x)$ denote the loss function. The first-order change under perturbation v is

$$L(x + v) = L(x) + \langle \nabla_x L(x), v \rangle + O(\|v\|^2).$$

Since $\nabla_x L(x) = J_h(x)^\top \nabla_h L(h(x))$, the gradient lies in the image of the Jacobian transpose and is therefore orthogonal (under g_x) to $\ker J_h(x)$. The loss-maximizing perturbation under any norm constraint therefore has nonzero projection onto $A_x = (\ker J_h(x))^\perp$, and under the pullback metric g_x the maximizer lies entirely in A_x . $\square$ $\square$

This theorem formalizes the intuition underlying gradient-based adversarial methods: the fast gradient sign method computes an approximation to the transverse direction, and the attack succeeds because the pullback metric concentrates sensitivity in the adversarial subspace. The singular foliation picture further explains the empirical clustering of adversarial examples near classification boundaries: these boundaries are precisely the singular strata of the foliation, where the transverse bundle expands abruptly.

Transferability also acquires a clean formulation. Two models S and T share adversarial directions when their foliations have similar leaf structures. CKA similarity is a proxy for foliation alignment: it measures whether the two models impose similar pairwise distances on the same inputs, which is equivalent to asking

whether their local equivalence classes are organized similarly. High CKA implies that the foliations are approximately aligned, so the transverse cones are similarly oriented and attacks transfer. Low CKA implies foliation divergence and near-orthogonal transverse cones.

The scaling behavior follows from dimensional considerations. As model size increases, the effective rank of $J_h(x)$ grows, the adversarial subspace expands in dimension, and independently trained models develop foliations whose orientations are less correlated. In high-dimensional spaces the expected overlap between two adversarial subspaces U_S and U_T scales roughly as

$$\mathbb{E} \left[\frac{\dim(U_S \cap U_T)}{\dim(U_S)} \right] \sim \frac{1}{d}.$$

Larger models therefore possess richer adversarial geometry but lower cross-model alignment, explaining the empirical observation that attacks become less transferable as model size grows, while each individual model remains vulnerable.

8 Representation Fields as Fiber Bundles and Grothendieck Fibrations

The representation map of a neural network can be interpreted naturally as a section of a fiber bundle over the input manifold, and this interpretation admits a canonical categorical formalization as a section of a Grothendieck fibration.

Let X denote the input space and let $E = X \times \mathbb{R}^d$ be the trivial bundle whose fibers represent the embedding space. The neural network determines a section

$$s : X \longrightarrow E, \quad s(x) = (x, h(x)).$$

Under this interpretation the learned representation is a field assigning to each point in input space a vector in the embedding fiber. The degeneracy structures discussed above correspond to singularities in the differential structure of this section: regions of semantic invariance correspond to directions along which s varies negligibly, while adversarial directions correspond to directions in which s exhibits large gradients.

The bundle interpretation makes it possible to interpret adversarial perturbations as failures of transversality between the section s and the classification boundary in the total space E . Training can be understood as shaping the section so that it aligns with the semantic foliation of the input space induced by the task. Robustness corresponds to maximizing the angle between the classification

boundary and the leaves of the foliation, measured in the metric induced by the section’s derivative.

The categorical formalization replaces the topological bundle with a Grothendieck fibration. Let $\mathcal{B}$ be a category whose objects represent input contexts—local regions $U \subseteq X$ together with admissible transformations within those regions—and whose morphisms $f : U \rightarrow V$ represent context transformations such as preprocessing operations, token substitutions, or small perturbations. Define the total category $\mathcal{E}$ whose objects are pairs (U, r) with $U \in \mathcal{B}$ and $r : U \rightarrow \mathbb{R}^d$ a representation assignment over that context. The projection functor $p(U, r) = U$ forgets the representation. For each morphism $f : U \rightarrow V$ and representation $r : V \rightarrow \mathbb{R}^d$, the pullback $f^*r = r \circ f$ satisfies the cartesian lifting property, making $p : \mathcal{E} \rightarrow \mathcal{B}$ a Grothendieck fibration.

The neural network determines a section $s : \mathcal{B} \rightarrow \mathcal{E}$ of this fibration, assigning to each context U the restricted representation $s(U) = (U, h|_U)$. By construction, $p \circ s = \text{id}_{\mathcal{B}}$.

Under this interpretation adversarial perturbations correspond to morphisms in $\mathcal{B}$ that induce large displacements in the fiber through the section s . The geometric problem of adversarial robustness therefore becomes a problem about how sections of a representation fibration behave under base category morphisms. This is precisely the categorical language in which the subsequent sections will describe dependent type theory: in both settings, objects vary over a base space of contexts, and coherent global behavior is described by sections of a fibration. The representation map of a neural network occupies the same formal role as a term inhabiting a dependent type—both are sections assigning values consistently across a space of contexts.

9 The λ -Cube as a Coordinate System for Constraint Spaces

The geometric language developed to describe neural representation spaces—manifolds, foliations, sections, bundles—is not merely metaphorical. The same mathematical structures arise in the categorical semantics of dependent type theory, and understanding this connection reveals a deeper unity between adversarial machine learning, formal logic, and geometry. The λ -cube provides the clearest entry point.

The λ -cube is traditionally introduced as a diagram classifying typed λ -calculi by their expressive power. From this perspective it appears to be primarily a taxonomy. This presentation obscures what the cube actually represents. At

the deepest level, the λ -cube is a coordinate system for a space of dependency relations between levels of abstraction. Any typed formal system involves at least three levels: *terms*, the executable expressions; *types*, which classify terms; and *kinds* (or universes), which classify types. The cube arises from asking a precise question about how these levels may depend on one another. Each of the three axes corresponds to one permitted cross-level dependency:

- *Types depending on terms.* This axis introduces dependent types. When a type $B(x)$ may mention a specific term $x : A$, the type acts not as a static predicate but as a constraint whose shape varies with the position in the base space. The typing judgment $x : A \vdash B(x) : *$ describes a family of types parameterized by elements of A .
- *Terms depending on types.* This axis introduces polymorphism. A term $\Lambda\alpha. t$ abstracts over a type variable, producing an expression that inhabits a type for every type substituted for α . Polymorphic terms are uniform sections across the entire universe of types.
- *Types depending on types.* This axis introduces type operators, allowing types to be constructed by applying higher-order type-level functions. A type operator $F : * \rightarrow *$ transforms one type into another; its existence requires that types themselves be treatable as objects manipulable by the formal language.

A formal system becomes self-describing when it can express objects, constraints on those objects, and constraints on those constraints. The λ -cube represents the minimal dependency lattice required for this reflexive structure. When no cross-level dependency is permitted, the result is $\lambda_{\rightarrow}$. Each additional axis adds one cross-level dependency. At the top vertex, where all three dependencies coexist, lies the Calculus of Constructions λC .

This interpretation of the cube as a coordinate system for dependency relations carries the key insight: the cube does not classify programming languages. It describes the minimal skeleton of cross-level dependency relations needed for a formal system to be self-describing. Any system capable of expressing objects, their constraints, and the constraints on those constraints must acquire structures equivalent to the three axes of the cube. The neural network context makes this resonance visible: neural networks are precisely systems whose internal representations describe constraints over input manifolds, and the λ -cube describes the minimal architecture for reasoning about such constraint hierarchies.

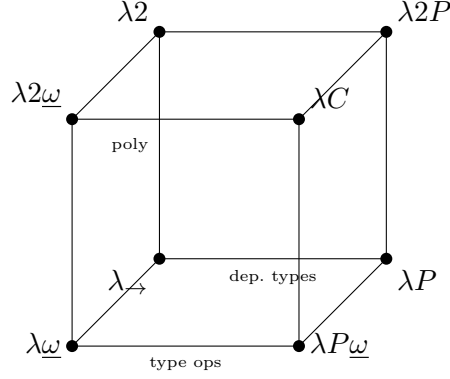


Figure 1: The λ -cube. Axes correspond to types depending on terms (depth), terms depending on types (height), and types depending on types (width). The Calculus of Constructions λC occupies the top vertex.

10 Comprehension Categories and the Emergence of the Cube

The fact that the λ -cube arises naturally from minimal categorical data is made precise by the theory of comprehension categories. A comprehension category provides the minimal structural account of the relationship between contexts, types, and terms, and from this account the cube's structure emerges automatically.

Definition 10.1 (Comprehension Category). A *comprehension category* consists of a fibration $p : \mathcal{E} \rightarrow \mathcal{C}$ together with, for every object A lying over Γ in $\mathcal{E}$, a chosen morphism

$$\chi_A : \Gamma.A \longrightarrow \Gamma$$

in $\mathcal{C}$, called the *context extension* map, with the property that the pullback of A along χ_A is the generic element of A .

The base category $\mathcal{C}$ represents contexts: its objects are environments $\Gamma = (x_1 : A_1, \dots, x_n : A_n)$, and its morphisms are substitutions between contexts. The total category $\mathcal{E}$ contains dependent types together with their context information. A type judgment $\Gamma \vdash A : *$ corresponds to an object $A \in \mathcal{E}$ with $p(A) = \Gamma$. The context extension $\Gamma.A$ is the context obtained by adding a fresh variable of type A ; the comprehension map χ_A expresses the projection forgetting that variable. Geometrically, $\Gamma.A$ behaves like the total space of a bundle whose base is Γ and whose fiber at each point consists of the possible values of A .

This minimal structure generates the fundamental operations of type theory through adjunctions. Given a substitution $f : \Delta \rightarrow \Gamma$, the fibration provides a reindexing functor $f^* : \mathcal{E}_\Gamma \rightarrow \mathcal{E}_\Delta$. Dependent products appear when this functor

possesses a right adjoint:

$$\Pi_f \dashv f^*.$$

The right adjoint Π_f maps a type B over Δ to a type $\Pi_f B$ over Γ that aggregates all coherent selections from B —the space of sections of the bundle. Dependent sums appear as the corresponding left adjoint:

$$f^* \dashv \Sigma_f,$$

constructing the total space by pairing each base element with an element of its fiber. When both adjoints exist for every projection in the fibration, the comprehension category models a locally cartesian closed category.

Once this structure is in place, the three axes of the λ -cube emerge automatically from the question of how many levels of the comprehension structure may serve as bases for further comprehension. Allowing types to serve as bases for further type families introduces the dependent-type axis. Allowing types to parameterize terms polymorphically introduces the polymorphism axis. Allowing types to depend on other types introduces the type-operator axis. The Calculus of Constructions arises when all three levels may serve as base spaces for fibrations and the comprehension structure is closed under all three forms of dependency.

The cube is therefore not a classification imposed from outside but a structure that grows organically from the minimal categorical data of a comprehension category. It is the dependency skeleton of any self-describing formal system.

11 Grothendieck Fibrations and the Geometry of Dependent Types

The connection between type theory and Grothendieck fibrations is where the geometric interpretation of logic becomes fully explicit.

Definition 11.1 (Grothendieck Fibration). A functor $p : \mathcal{E} \rightarrow \mathcal{B}$ is a *Grothendieck fibration* if for every object $e \in \mathcal{E}$ and every morphism $f : b \rightarrow p(e)$ in $\mathcal{B}$, there exists a *cartesian lifting*: a morphism $\bar{f} : \bar{b} \rightarrow e$ in $\mathcal{E}$ with $p(\bar{f}) = f$, through which every other morphism over f factors uniquely.

The base category $\mathcal{B}$ represents contexts. For each $\Gamma \in \mathcal{B}$, the fiber category $\mathcal{E}_\Gamma$ is the category of types in context Γ . The correspondence with type-theoretic

judgments is nearly perfect:

$$\begin{aligned}\Gamma \in \mathcal{B} &\longleftrightarrow \text{a context} \\ A \in \mathcal{E}_\Gamma &\longleftrightarrow \Gamma \vdash A : * \\ t \in \mathcal{E}_\Gamma(A, B) &\longleftrightarrow \Gamma \vdash t : A \rightarrow B\end{aligned}$$

A morphism $f : \Delta \rightarrow \Gamma$ in the base induces a reindexing functor $f^* : \mathcal{E}_\Gamma \rightarrow \mathcal{E}_\Delta$ via cartesian lifting. This is substitution.

The dependent product $\Pi_{x:A} B(x)$ arises as the right adjoint of reindexing along the projection $p_A : \Gamma.A \rightarrow \Gamma$:

$$\mathcal{E}_{\Gamma.A} \begin{array}{c} \xrightarrow{\Pi_{p_A}} \\ \xleftarrow{p_A^*} \end{array} \mathcal{E}_\Gamma$$

The right adjoint assembles all coherent sections of the bundle into a single object. In logical terms this is universal quantification; in geometric terms it is the space of sections of a fiber bundle. The dependent sum $\Sigma_{x:A} B(x)$ appears as the left adjoint, constructing the total space.

Under sheaf semantics this structure acquires additional geometric content. A context acts like an open region of a base space, a type describes how objects vary across that region, and a term is a consistent section. Logical proof becomes the construction of global sections from local constraints. The subject-reduction property of typed calculi—the fact that evaluation preserves typing—is the condition that the evaluation flow remains within the fibers of the fibration, never leaving the constraint manifold defined by the type.

12 The λ -Cube as a Lattice of Adjunctions

The deepest way to understand the λ -cube is through adjoint functors. Each axis of the cube corresponds to a distinct adjoint pair becoming available inside the internal language.

The simply typed λ -calculus $\lambda_{\rightarrow}$ lives in a cartesian closed category. The function type $A \rightarrow B$ is the exponential B^A , and the currying isomorphism

$$\text{Hom}(X \times A, B) \cong \text{Hom}(X, B^A)$$

expresses the adjunction $(- \times A) \dashv (-)^A$. The λ -abstraction rule is the internal language of this adjunction.

The dependent-type axis introduces, for each projection $p_A : \Gamma.A \rightarrow \Gamma$, the

triple adjunction

$$\Sigma_{p_A} \dashv p_A^* \dashv \Pi_{p_A}.$$

This corresponds to the categorical transition from cartesian closed categories to locally cartesian closed categories. The polymorphism axis introduces the ability to quantify over the universe of types, requiring the universe to be treated as an object and polymorphic terms to correspond to natural transformations. The type-operator axis introduces internal hom-objects at the level of types, adding exponentials between kinds.

The Calculus of Constructions λC occupies the vertex where all three adjoint structures coexist. The Curry–Howard–Lambek correspondence makes the full triangulation explicit:

Logic	Type Theory	Category Theory
Propositions	Types	Objects
Proofs	Terms	Morphisms
Universal quantification	Π -type	Right adjoint to reindexing
Existential quantification	Σ -type	Left adjoint to reindexing
Implication	Function type	Exponential / internal hom
Substitution	β -reduction	Adjoint transpose

The adjoint lattice extends naturally beyond three dimensions. Modal type theories arise from adjunctions between modalities and inclusion functors. Linear type theories arise from adjunctions between tensor products and linear internal homs. Universe hierarchies $\text{Type}_0 : \text{Type}_1 : \text{Type}_2 : \dots$ arise from adjunctions between universe levels. The λ -cube is the three-dimensional shadow of this larger structure: the most visible cross-section because it is the minimal one containing all three fundamental modes of dependency.

13 Type Theory as Information Geometry

Once the λ -cube is interpreted geometrically as a space of constraints and flows, its structure becomes comparable to the manifolds studied in information geometry and statistical field theory. The comparison reveals that typed computation and statistical inference share a common geometric skeleton.

In information geometry, a statistical model is a Riemannian manifold whose coordinates parameterize probability distributions. A point corresponds to a particular distribution; a curve corresponds to a continuous transformation of that distribution. The Fisher information metric equips the manifold with an intrinsic

notion of distance, and divergence measures define the geodesic structure. Learning and inference are motion through this manifold.

The typing judgment $\Gamma \vdash t : T$ can be interpreted in a parallel way. The context Γ specifies a coordinate system; T defines a constraint surface within the space of program states; and the judgment asserts that t lies on that surface. Beta reduction in the λ -calculus is a dynamical flow on this space: each reduction step transforms a term while preserving its type, so the flow remains on the constraint manifold. The subject-reduction theorem is the condition that the flow never leaves the manifold.

Dependent types introduce fiber bundle structure. A family $x : A \vdash B(x)$ defines a bundle over A whose fiber at each a is the constraint manifold $B(a)$. A dependent function of type $\Pi_{x:A} B(x)$ is a section. This is the exact structure of a physical field: a field assigns to each spacetime event an element of a fiber determined by the gauge structure. The Π -type performs the same role as integration of a field over a base space.

The λ -cube describes which kinds of fields over constraint manifolds are permitted. The dependent-type axis allows the constraint manifold to vary with the base point. The polymorphism axis allows programs to be uniform across the entire universe of manifolds. The type-operator axis allows manifolds to depend on other manifolds. The Calculus of Constructions permits fields over fields over fields, allowing constraints, programs, and meta-structures to interact recursively. In information-geometric terms, the system can describe statistical models over statistical models—manifolds parameterized by manifolds.

14 Homotopy Type Theory and the Extension Beyond the Cube

Homotopy type theory (HoTT) demonstrates that the geometric interpretation of type theory can be pushed to the point where the logic of type theory and the logic of homotopy theory become provably equivalent. In HoTT, types are topological spaces, elements are points, and equality proofs are paths. The identity type $\text{Id}_A(a, b)$ is the space of paths from a to b in A . Higher identity types correspond to surfaces connecting paths, and so on.

Under this interpretation, dependent types become genuine fibrations in the topological sense. The family $x : A \vdash B(x)$ is a continuous map from the total space $\Sigma_{x:A} B(x)$ to the base A with the standard path-lifting properties. The Π -type is the space of sections of this fibration.

The univalence axiom asserts that equivalent types are identical. In categorical terms, the identity type of the universe is homotopy equivalent to the space of type equivalences. The universe therefore carries the structure of a classifying space: paths are equivalences, surfaces are natural isomorphisms, and higher-dimensional cells are higher coherences.

Higher topos theory provides the categorical framework for this extension. An $(\infty, 1)$ -topos is a category whose internal language is precisely HoTT. The λ -cube describes the internal language of an ordinary topos—the 0-dimensional case—while HoTT describes the internal language of an $(\infty, 1)$ -topos. Between these extremes lie the internal languages of n -toposes for each natural number n , forming an infinite hierarchy extending the cube in a direction orthogonal to all three of its original axes.

Derived algebraic geometry extends this picture further. The λ -cube therefore appears, in the longest view, as the minimal coordinate patch of an infinite-dimensional space of type theories parameterized by their homotopical dimension and homological depth.

15 Representation Learning as a Renormalization Flow

The geometric interpretation of neural representation spaces admits a natural analogy with renormalization group (RG) flows in statistical physics. In both settings a complex high-dimensional system is described at multiple levels of resolution, and the effective behavior observed at a given scale depends on how fine-grained degrees of freedom have been integrated out.

Let $h_\theta : X \rightarrow \mathbb{R}^d$ denote the representation map parameterized by θ . Training modifies θ through gradient descent, producing a trajectory

$$\theta_0 \longrightarrow \theta_1 \longrightarrow \dots$$

that induces a corresponding evolution of representation geometries. At early stages of training the representation map tends to preserve fine-grained variations in the input. As training progresses, the model learns to collapse directions irrelevant to the task—the small-singular-value directions discussed in the spectral analysis above. This collapse is directly analogous to the separation of relevant and irrelevant operators under renormalization group flow. Relevant operators dominate the large-scale behavior; irrelevant operators decay under coarse-graining.

The foliation structure of representation space emerges as a fixed point of this

renormalization-like process. Directions associated with small singular values form the tangent bundle of the semantic equivalence foliation. Directions associated with large singular values define the transverse bundle along which classification changes occur. Training flows the system toward a configuration in which this spectral separation is maximized, just as RG flows drive a system toward a fixed point by integrating out high-frequency fluctuations.

This perspective also clarifies the scaling behavior of adversarial transferability. As model size increases, the dimensionality of the representation manifold grows and the number of relevant directions increases. However, independently trained models tend to develop different orientations for their relevant subspaces. In high-dimensional spaces the expected overlap between two adversarial subspaces U_S and U_T scales as $1/d$, so larger models possess richer adversarial geometry but lower cross-model alignment. The renormalization interpretation suggests that large models develop non-renormalizable degrees of freedom—gradient directions that cannot be described by a simple shared effective theory—which is why their adversarial directions decorrelate across independently trained instances.

Within the broader framework developed in this essay, this renormalized representation manifold functions as a constraint field over the space of inputs. Neural inference corresponds to evaluating sections of this field; adversarial perturbations correspond to trajectories that leave the local constraint leaf and move along unstable transverse directions. Seen from this perspective, representation learning, dependent type theory, and categorical semantics all describe systems in which structures evolve under constraint-preserving flows. The renormalization analogy provides a physical intuition for the essay’s central claim: intelligence, logic, and learning are all governed by the geometry of constraint fields defined over spaces of contexts.

16 Semantic Manifolds and the Emergence of Adversarial Cones

The geometric framework developed throughout this essay allows gradient-based adversarial attacks to be interpreted as natural differential-geometric operations on the representation manifold: they are first-order approximations to geodesic escape from the leaves of the semantic foliation.

Let $F = f \circ h : X \rightarrow \mathbb{R}^k$ be the composed classifier and let $\mathcal{F}$ be the foliation induced by training. At each point x , the tangent space $T_x\mathcal{F}$ consists of directions that leave the classification invariant to first order, while the transverse bundle

$A_x = (T_x \mathcal{F})^\perp$ contains the adversarially sensitive directions (Theorem 7.3).

A gradient-based adversarial attack computes the direction of steepest ascent of the loss under a norm constraint. The sign perturbation

$$\delta^* = \epsilon \operatorname{sign}(\nabla_x L(x))$$

approximates the projection of $\nabla_x L(x)$ onto the ℓ_∞ unit sphere. Since $\nabla_x L(x) \in A_x$ (it is orthogonal to the leaf tangent space), this perturbation moves the input in the adversarially sensitive direction: the transverse escape direction of the foliation.

The set of all such directions at x forms the *adversarial cone*:

$$C_x = \{v \in T_x X \mid \langle \nabla_x L(x), v \rangle > 0\}.$$

In high-dimensional spaces this half-space occupies a large fraction of the ambient tangent space. Consequently, randomly sampled perturbations have non-negligible probability of falling within C_x , which explains the empirical abundance of adversarial examples.

The geometry becomes particularly striking near singular strata, where $\dim T_x \mathcal{F}$ decreases discontinuously. At such points the transverse bundle expands abruptly and the adversarial cone widens. These are precisely the unstable classification boundaries at which adversarial examples cluster most densely. Robustness is therefore equivalent to increasing the geodesic distance from typical data points to the nearest singular stratum of the foliation: a network is adversarially robust insofar as a significant geodesic displacement is required to escape the local leaf.

The connection to adversarial transfer is now geometric and explicit. Two models S and T have similar adversarial cones when their foliation tangent bundles are aligned. CKA measures this alignment. A high-CKA surrogate shares the foliation geometry of the target, so attacks computed by following $\nabla_x L_S(x)$ lie in the adversarial cone of T as well. A low-CKA surrogate has rotated adversarial cones, and transfer fails because $\nabla_x L_S(x)$ is orthogonal to the sensitive directions of T . The Cox–Bunzel protocol, by combining both families of surrogates, covers adversarial cones from multiple orientations and thereby provides representative coverage of the full transverse bundle of the target.

17 Geometric Summary Theorem

The analysis developed throughout this essay can be unified into a single formal statement that makes the central claim of the paper precise.

Theorem 17.1 (Geometry of Constraints). *Let $\mathfrak{S}$ denote any of the following three classes of system:*

- (i) *A trained neural network with representation map $h : X \rightarrow \mathbb{R}^d$;*
- (ii) *A dependent type theory presented as a comprehension category $p : \mathcal{E} \rightarrow \mathcal{C}$ with context extension;*
- (iii) *A family of statistical models presented as a fibered manifold over a base of parameters.*

In each case there exists a structure of the following kind:

- (a) *A base space of contexts B , whose points parameterize the local conditions under which the system operates.*
- (b) *A constraint field over B : a Grothendieck fibration $p : \mathcal{E} \rightarrow B$ whose fiber $\mathcal{E}_b$ over context b is the space of admissible states or types under condition b .*
- (c) *A canonical section $s : B \rightarrow \mathcal{E}$ describing the globally coherent behavior of the system, satisfying $p \circ s = \text{id}_B$.*
- (d) *A foliation $\mathcal{F}$ of the total space $\mathcal{E}$ into leaves of approximate equivalence, induced by the task structure or type constraints. The tangent bundle $T\mathcal{F}$ encodes the directions along which the system’s output is invariant; the transverse bundle $T\mathcal{E}/T\mathcal{F}$ encodes the directions of sensitivity.*
- (e) *An instability locus: the singular strata of $\mathcal{F}$, where the leaf dimension drops and the transverse bundle expands. In neural networks these are the adversarial boundaries; in type theory these are the boundaries between normal and stuck computation; in information geometry these are the boundaries of statistical distinguishability.*

The system is robust insofar as the section s avoids the instability locus: the geodesic distance from $s(b)$ to the nearest singular stratum, measured in the pull-back metric induced by the Jacobian of the section, is large. The characteristic failure modes of each domain—adversarial perturbations, type errors, and statistical non-identifiability—correspond to different mechanisms by which the section approaches or crosses a singular stratum.

Moreover, when two systems of the same class are compared, their structural similarity (measured by CKA in neural networks, by natural isomorphism of fiber categories in type theory, or by statistical distance of the corresponding fibered manifolds) predicts the degree to which the transverse bundles of their respective

sections are aligned, and hence the degree to which instabilities in one system transfer to the other.

Proof sketch. Parts (a)–(c) hold for neural networks by the fiber bundle construction of Section 8 and the Grothendieck fibration formalization in the appendix. Parts (a)–(c) hold for comprehension categories by definition and for statistical manifolds by the Grothendieck construction applied to the functor assigning fiber manifolds to base parameters.

Part (d) holds for neural networks by the spectral analysis of the Jacobian (the foliation is induced by the kernel of $D(f \circ h)$) and for comprehension categories by the fibered structure of the type theory (the foliation is induced by definitional equality in the fibers). Part (e) is Theorem 7.3 in the neural setting and corresponds to non-normal-form terms in the type-theoretic setting.

Robustness follows from the geodesic interpretation of adversarial attacks in Section 16: a perturbation produces a singular crossing if and only if the geodesic distance in the pullback metric from the current section value to the instability locus is less than the perturbation radius. Structural similarity and transferability are related by the foliation alignment argument of Section 7: aligned foliations have aligned transverse bundles, and perturbations effective in one system lie in the adversarial cone of the other. $\square$ $\square$

Remark 17.2. Theorem 17.1 should be understood as a structural theorem rather than a quantitative one: it identifies the common architecture underlying three domains without specifying uniform quantitative bounds across them. The quantitative theory specific to each domain (NP-hardness results for adversarial verification, coherence conditions for comprehension categories, and Fisher information bounds for statistical distinguishability) is richer than what a single unified theorem can encode. The theorem’s value lies in identifying the shared geometric vocabulary that makes each domain’s quantitative results interpretable in the others.

18 Returning to Machine Learning: Constraint Fields in Practice

With the full theoretical framework in place it is possible to revisit the applied problem of adversarial robustness and state precisely what the geometric theory contributes to its solution.

A trained neural network defines a representation field over its input space. At each input x , the network’s internal state $h_M(x) \in \mathbb{R}^d$ is the value that the

representation field assigns to x . The training process shapes this field to assign similar values to semantically similar inputs, encoding the task geometry in the structure of the field.

Under the fibration interpretation established in Theorem 17.1, the network’s representation is a section of a bundle over input space whose fiber-smoothness properties determine the model’s adversarial behavior. CKA-based representational similarity is a probe of fibrational alignment between two networks: high CKA indicates that the respective sections organize the input manifold into similar geometric patterns, so the adversarial cones are aligned and attacks transfer. The Cox–Bunzel testing protocol is a strategy for obtaining representative coverage of the adversarial cone by sampling surrogates from both aligned and orthogonal orientations.

The renormalization interpretation of Section 15 suggests that the most robust possible network is one whose representation field has been trained to its fixed-point configuration: one in which the semantic foliation is maximally sharp (large spectral gap in the Jacobian singular values) and maximally stable (the instability locus is far from typical data points in the pullback metric). This corresponds to a model whose constraint field is strongly laminar—varying rapidly in a small number of task-relevant directions and negligibly in all others—so that adversarial perturbations require a large geodesic step to reach the instability locus.

The challenge of AI robustness is, at its deepest level, the challenge of understanding and controlling the geometry of representation spaces as constraint fields over the manifold of possible inputs. The techniques developed for this problem—CKA-based representational similarity, surrogate partitioning, diagonal box metrics—are instruments for probing the geometry of learned constraint fields. The geometric framework of this essay provides the conceptual vocabulary in which those instruments can be combined into a coherent theory.

19 Convergence: The Geometry of Constraints

This essay has traced a path from adversarial examples through representation geometry, foliation theory, Jacobian spectral analysis, transformer attention as a dynamic metric, categorical semantics, Grothendieck fibrations, comprehension categories, the λ -cube and its adjoint lattice, information geometry, homotopy type theory, renormalization flows, and the derivation of adversarial cones as geodesic escape directions. The diversity of these topics conceals a structural unity that the foregoing analysis has progressively revealed and Theorem 17.1 has formalized.

The unity can be stated as a single organizing claim: complex systems—

whether machine learning models, typed programming languages, or mathematical proof systems—are best understood as layered spaces of states, constraints, and meta-constraints, and the phenomena characteristic of each domain are instances of motion through such layered spaces under the governance of their constraint geometry.

In machine learning, states are input representations; constraints are the typing-like conditions imposed by training objectives and architectural inductive biases; adversarial vulnerability is the existence of transverse directions along which small motion produces large change in behavioral category; and robustness evaluation is the problem of characterizing the geometry of these transverse directions without complete knowledge of the full constraint field.

In type theory, states are program terms; constraints are types (constraint manifolds in the state space); computation is motion along those manifolds governed by reduction rules; and the λ -cube describes the minimal configuration of cross-level dependencies needed for a language to reason about itself.

In categorical semantics, the same structure appears as a Grothendieck fibration: a functor from a total category of constrained objects to a base category of contexts, equipped with reindexing functors and their adjoints. The fibration encodes, in categorical language, exactly the structure of substitution, dependent quantification, and context extension that type theory requires syntactically.

In information geometry, the structure appears as a family of statistical manifolds parameterized by a base space of models, with sections corresponding to consistent statistical inferences across the family.

What appears initially as a collection of unrelated technical tools—adversarial testing frameworks, dependent type systems, categorical fibrations, information geometry—is, under the analysis developed here, a collection of partial descriptions of the same underlying mathematical object: a geometry of constraints governing how structures vary across contexts and interact across levels of abstraction. The λ -cube, Grothendieck fibrations, and neural representation manifolds are, each in their own register, charts for different regions of this geometry. The enterprise of understanding any one of them in depth is, in the end, the same enterprise as understanding all the others.

A Neural Representation Maps as Sections of a Grothendieck Fibration

This appendix makes explicit the categorical construction promised in Section 8. The goal is to demonstrate, without appeal to metaphor, that a neural representation map is a section of a Grothendieck fibration over input contexts in the same formal sense that a dependent-type term is a section of the comprehension fibration.

Construction of the Base Category

Let X denote the input space of a neural network. Define a category $\mathcal{B}$ whose objects are *input contexts*: pairs $(U, \mathcal{T})$ where $U \subseteq X$ is a measurable region and $\mathcal{T}$ is a set of admissible transformations of U (preprocessing operations, token substitutions, perturbations of bounded norm). A morphism

$$f : (U, \mathcal{T}) \longrightarrow (V, \mathcal{T}')$$

is an admissible map $f : U \rightarrow V$ that intertwines the transformation structures: $\mathcal{T}' \circ f \subseteq f \circ \mathcal{T}$. Composition is ordinary function composition; identity morphisms are inclusions. The category $\mathcal{B}$ captures the space of contexts over which the network operates.

Construction of the Total Category and Projection

Define the total category $\mathcal{E}$ whose objects are triples $(U, \mathcal{T}, r)$ with $(U, \mathcal{T}) \in \mathcal{B}$ and $r : U \rightarrow \mathbb{R}^d$ a measurable representation assignment over that context. A morphism

$$(U, \mathcal{T}, r) \longrightarrow (V, \mathcal{T}', r')$$

consists of a base morphism $f : (U, \mathcal{T}) \rightarrow (V, \mathcal{T}')$ together with a commutation condition: $r = r' \circ f$ (the representation over U is the pullback of the representation over V along f). The projection functor $p : \mathcal{E} \rightarrow \mathcal{B}$ is defined by

$$p(U, \mathcal{T}, r) = (U, \mathcal{T}), \quad p(f, \text{comm.}) = f.$$

Verification of the Fibration Property

For each object $(U, \mathcal{T}, r')$ of $\mathcal{E}$ and each morphism $f : (U, \mathcal{T}) \rightarrow (V, \mathcal{T}')$ in $\mathcal{B}$, define the pullback representation

$$f^*r' = r' \circ f : U \longrightarrow \mathbb{R}^d.$$

The morphism $(U, \mathcal{T}, f^*r') \rightarrow (V, \mathcal{T}', r')$ consisting of $(f, \text{id}_{r' \circ f})$ is cartesian: every morphism $(W, \mathcal{T}'', r'') \rightarrow (V, \mathcal{T}', r')$ in $\mathcal{E}$ lying over a morphism $g : (W, \mathcal{T}'') \rightarrow (U, \mathcal{T})$ in $\mathcal{B}$ factors uniquely through the pullback. This is precisely the universal property of cartesian liftings. The functor p is therefore a Grothendieck fibration.

The Neural Network as a Section

The neural network with representation map $h : X \rightarrow \mathbb{R}^d$ determines a section $s : \mathcal{B} \rightarrow \mathcal{E}$ of this fibration:

$$s(U, \mathcal{T}) = (U, \mathcal{T}, h|_U), \quad s(f) = (f, h|_U = h|_V \circ f).$$

By construction $p \circ s = \text{id}_{\mathcal{B}}$, confirming that s is a section. The commutativity condition $h|_U = h|_V \circ f$ holds automatically for any admissible f , because h is defined globally on X and $h|_U$ is simply its restriction. The section is therefore well-defined.

Adversarial Perturbations as Base Morphisms

An adversarial perturbation of bounded norm ϵ at input $x \in U$ defines a morphism in $\mathcal{B}$:

$$\delta : (U, \mathcal{T}) \longrightarrow (U_\epsilon, \mathcal{T}_\epsilon)$$

where U_ϵ is the ϵ -ball around x and δ maps x to $x + v$ for some v with $\|v\| \leq \epsilon$. The total displacement induced by s along this morphism is

$$s(U_\epsilon, \mathcal{T}_\epsilon) = (U_\epsilon, \mathcal{T}_\epsilon, h|_{U_\epsilon}).$$

The adversarial perturbation is effective when $h(x + v) - h(x)$ is large relative to $\|v\|$, which is equivalent to saying that the section s has a large derivative in the direction v : the morphism δ induces a large displacement in the fiber $\mathbb{R}^d$ through the section. This is precisely the condition—section gradient large in the transverse direction—identified in Theorem 7.3.

Interpretation

This construction demonstrates that the neural network’s representation map occupies the same formal role as a term inhabiting a dependent type: both are sections of Grothendieck fibrations assigning values consistently across a space of contexts. The base category of input contexts corresponds to the base category of typing contexts; the fiber of representation assignments corresponds to the fiber of type-admissible terms; the pullback reindexing corresponds to substitution; and adversarial perturbations correspond to context morphisms that produce large fiber displacements through the section. The geometric problem of adversarial robustness—controlling how sections behave under base category morphisms—is formally identical to the problem of ensuring that term-evaluation is stable under context change, which is the subject-reduction problem of type theory.

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